

What Do We Use Robots For?

Student Edition — VEX EXP • C++ in VS Code

Name: _____ Date: _____ Period: _____

Introduction

Robots give us access to places people cannot safely go, protect us from danger, and take over tedious or repetitive work — the **three Ds**: dirty, dangerous, and dull jobs. Some of the first robots handled radioactive materials in the 1940s; today robots weld and assemble in factories, assist in surgery, vacuum homes, and explore other planets. Every robotic system has five main parts: a **controller** (its brain), a **program** that tells it what to do, a **manipulator** that provides movement, an **end effector** that acts like a hand, and a **power supply**. In this activity, your team researches one class of robot and teaches the class about it. Read Chapter 2 in the textbook for the full explanation.

Learning Objectives

- Describe what a robot is and the five parts of a robotic system.
- Research a category of robot and explain the task it performs and the human function it replaces or extends.
- Discuss benefits, drawbacks, and societal impacts of automation, including effects on jobs.
- Present findings clearly and summarize peers' presentations.

Key Ideas (quick reference)

- **A robot senses → decides → acts:** it senses its surroundings, decides using its program, acts on the world, then checks again. A machine that never senses (a toaster) is not a robot.
- **Five parts of a robotic system:** controller (brain), program (instructions), manipulator (movement), end effector (hand), power supply. On your EXP robot: Brain, your C++ program, chassis/arm, claw, EXP Battery.
- **Degrees of freedom & work envelope:** each flexible joint adds a direction of movement; the work envelope is the space the robot can reach — and the space people must stay out of.
- **Sensors close the loop:** touch (Bumper Switch), distance (Distance Sensor), light/color (Optical Sensor), motion (Inertial Sensor) — your kit has the same sensor families big robots use.

Equipment & Materials

- Engineering notebook and pencil
- Computer with internet access and presentation software
- Reference books or articles on robotics

Procedure

1. **Form teams and claim a category.** Your teacher assigns each team a robot category: industrial, medical, automated vehicles, household, military, toys, exploratory, rescue, or assistive.
2. **Choose a specific robot and research it** against every question in the research checklist below. Split questions among teammates, but review everything together.
3. **Build a short presentation** with sketches, photos, or video of the robot in action.

4. **Present, listen, and summarize.** Present to the class; take notes during other teams' presentations; then write a paragraph summarizing what you learned across all the robots.

Research Checklist (answer all nine)

1. What task does the robot perform, and what human function does it simulate, replace, or extend?
2. Where is it used, and what is its work envelope — how many degrees of freedom or flexible joints does it have?
3. Is its end effector multi-functional? If so, what else can it do?
4. How is the robot taught to perform its task?
5. What sensors does it have, and how does it use them?
6. What are the advantages and disadvantages of using a robot for this task?
7. What impact has it had — or could it have — on the people it serves?
8. What jobs and careers does this robot create?
9. How might this robot be altered to do more, or different, tasks in the future?

How you'll be graded: content (accurate, answers every checklist question), organization (easy to follow), delivery (on topic, audible, eye contact), and cooperation & teamwork. A beautiful presentation that skips three questions loses to a plain one that nails all nine.

Conclusion Questions

Answer in complete sentences.

1. Name the five main parts of a robotic system, and identify where each one is on your EXP robot.
2. In your own words, what is a work envelope, and why does it matter for safety around an industrial robot?
3. Which robot do you think will have the most significant impact on humans, and why?
4. What concerns do you have about the increasing use of robots in society?
5. What do you think about giving up control to a machine?